L1a: Course Orientation, Markets, and Financial Decisions

CHEME 5660 Cornell University

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Training only. This content is for training and informational purposes only. The teaching team does not make, give, or endorse any offer or solicitation to buy or sell securities or derivative products, or provide investment or trading advice or strategy.

Risk. Trading involves risk and is generally not recommended for those with limited resources, experience, or a low risk tolerance. Past performance does not guarantee future success.

Responsibility. You are responsible for your investment decisions based on your financial circumstances, objectives, risk tolerance, and liquidity needs.

Objectives for Today

Today we answer three questions: How does the course work? Why do financial markets matter even if you never become a trader? What makes one investment preferable to another?

Today's concepts

- Course roadmap: four units move from reading payments to making and testing financial decisions.
- Why markets matter: most adults and Cornell itself are investors, while companies use public markets to raise money.
- Our first decision problem: two investments pay different amounts at different times. How can we compare them?

Note: this lecture has no notebook and no notes packet. These slides are the record.

How This Course Runs

Across twenty-eight lectures, we'll move from understanding financial assets to valuing them, combining them into portfolios, and making investment decisions.

MATERIALS

Download each week from the repository releases page, or clone the repository. Released files stay frozen; corrections are announced.

TOOLS

Use Julia 1.12 through juliaup, Anaconda for Jupyter, and VS Code with the Julia extension. Set these up once.

QUESTIONS

Ask on the course discussion board rather than by email, so each answer reaches everyone with the same question.

Final project: automated trading with live data and live market access through the Alpaca Markets application programming interface.

Policies: grading, attendance, academic integrity, and accommodations are posted in full

Markets Connect You, Cornell, and Companies

YOUR SIDE (2025)

61%

of U.S. adults had a retirement account.

Keeping the default mix is still a choice.

That describes today's adults, not your future.

CORNELL'S SIDE (2025)

$10.9B

FY2025 endowment

$498M went into operations, mainly from endowment payout.

About 29% ($3.16B) of Cornell's endowment supported financial aid through its annual payout.

COMPANY SIDE

315

corporate IPOs in U.S. markets January 2025 through June 2026

An IPO lists a private company's shares on a public market.

Raise capital by selling new shares; **create liquidity** by letting shareholders trade.

Sources: Fed SHED, 2025; Cornell FY2025 report; Cornell activities; SEC IPO data

U.S. Markets Move About $1.85 Trillion a Day

To compare annual and daily figures, the daily column divides by 252 trading days. Industry rows show revenue; market rows show cash paid when securities trade.

	Annual total (USD)	Per trading day (USD)
Global biopharma	666B	2.64B
Global batteries	181B	0.72B
Together	847B	3.36B
US debt	304.7T	1.21T
US stocks	153.2T	608B
Prices paid for options	9.3T	37B
US markets together	467.2T	1.85T

Whether measured per year or per trading day, the market total is about 550 times the two-industry total.

Sources: market sizes 2025 (Fortune Business Insights); turnover 2024--2026 (SIFMA, Cboe)

Federal Deficits Require Treasury Borrowing

A federal deficit is financed by issuing Treasury debt.

FEDERAL DEBT | AUG. 20, 2026

$40.03T

about **$117,000 per U.S. resident**

Held by the public	**$32.28T**
Intragovernmental	**$7.75T**

LIQUIDITY-SUPPORT BUYBACK | AUG. 20

Securities offered	**$10.16B**
Announced maximum	**$4.00B**
Treasury accepted	**$1.86B**

The buyback retired portions of three older issues. It equaled 0.006% of publicly held debt and supported market liquidity.

Why elected officials care: continuing deficits require repeated borrowing from investors at market interest rates.

Note: Scale comparison, not a bill. Sources: Treasury debt; buyback; Census population.

Gross Notional Value Is About 108 Times Premium Paid

In 2025, 60.4 million option contracts were traded each day. Buyers paid USD 37B in premiums for about USD 4T in gross notional value:

USD 4T	/	USD 37B	=	108×
gross notional value		premium		$108 gross notional per $1 paid

Interestingly, about USD 3T of that value came from cash-settled S&P 500 index options. For an option, gross notional divided by premium approximates exposure-based leverage when $|\delta| \approx 1$.

Contract size: equity options usually cover 100 shares. For index options, the multiplier converts index points into dollars (SPX: $100 per point).

The Course Moves from Valuing Assets to Making Decisions

The first three units explore how to value debt, equity, and derivatives. The fourth unit uses these assets to build and test financial strategies.

One question connects all four units: what should you do?

Unit 1	Unit 2	Unit 3	Unit 4
Debt	**Equity**	**Derivatives**	**Decisions**
Treasury securities	Stocks and portfolios	Options	Financial strategies
weeks 1–2	weeks 3–7	weeks 8–11	weeks 12–15
compare payments across dates	build portfolios under uncertainty	value investments tied to other values	build and test financial strategies

weeks

This week starts with debt in the form of United States Treasury securities.

Every Unit Turns Information into a Choice

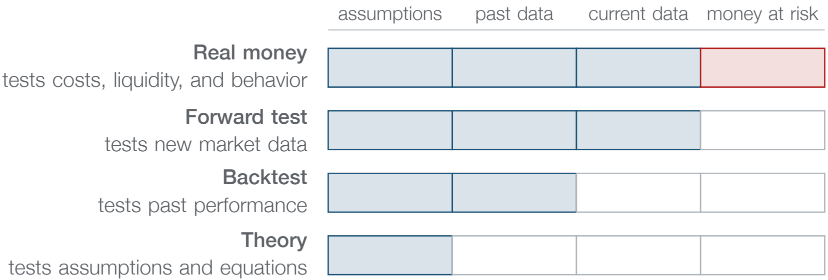
The asset changes, but the questions stay the same.

Unit	Observe	Choose	Evaluate
Debt	rates and payments	bond and maturity	price and rate risk
Equity	cash flows and growth	stock or portfolio	value and risk
Derivatives	price and volatility	contract and position	payoff and risk
Strategies	signals and limits	allocation rule	performance over time

Unit 4 combines the first three units into a strategy.

Strategy Validation Uses Four Levels of Evidence

Units 1 to 3 build models of prices and payoffs. Unit 4 evaluates strategies using past data, current data, and actual trades.



Each row includes the evidence below it and adds one new source.

Five Investments Have Different Payment Schedules

Suppose you could buy each set of future payments for $10,000 today. Start with the information shown: compare the amount and timing of the payments.

A	$10,028 back, four weeks from today
B	$236 every six months for 10 years, then $10,000
C	$264 every six months for **30** years, then $10,000
D	$271 every six months for 10 years, then $10,000
E	$121 every six months for 10 years, plus an inflation adjustment on every payment and on the $10,000

Which promise would you choose? Let's brainstorm and vote.

Rates: published rates for 2026-08-18, each scaled to a $10,000 outlay

Each Payment Schedule Represents a Different Security

	What it is called	Annual rate
A	4-week Treasury bill	3.63%
B	10-year Treasury note	4.71%
C	30-year Treasury bond	5.28%
D	investment-grade corporate bonds	5.42%
E	10-year inflation-indexed Treasury (TIPS)	2.41% real

Compare the amount, timing, and uncertainty of the payments. The schedule gives the amount and timing. The security type identifies which sources of uncertainty matter.

Careful: D is an index of many investment-grade corporate bonds, not one bond. E's rate is stated after inflation, unlike the others.

Compare the Amount, Timing, and Uncertainty of Payments

The schedules show amount and timing. The following questions identify three sources of uncertainty.

- C **What if you need the money before year 30?** Selling early means accepting the market price then, and that price changes with interest rates. Week 2.
- D **Why does it pay more?** Its rate is 5.42%, versus 4.71% for the ten-year Treasury. That gap reflects the chance of missed payments and the difficulty of selling quickly. Units 2 and 3.
- E **What does inflation protection change?** Its payments rise with inflation, so its 2.41% rate is not directly comparable with the other rates. Later in the course.

Amount and timing are not sufficient without these sources of uncertainty.

A Bank Can Fail Even When Its Securities Do Not Default

$200B
assets in March 2023

A bank with roughly this much in assets failed in two days. Its portfolio included one of the five kinds of promises on your list.

Which one, and why?

The security did not default; every payment that came due was made. Yet the bank still failed.

As you watch, identify which promise the bank held and what changed before it failed. Week 2 gives us a model for checking the explanation.

Clip: PBS NewsHour, 28 March 2023, 6 minutes

Summary

Financial decisions start with payment schedules, but the largest payment is not automatically the best choice.

Key takeaways

- **Markets connect investors, companies, and governments.** Cornell invests its endowment; companies issue shares; Treasury borrows when federal spending exceeds revenue.
- **A product name is a label.** What matters first is the payment schedule: what gets paid, when, and under what conditions.
- **Models earn trust one test at a time.** Theory, backtest, forward test, and real money use different evidence and can reveal different failures.

On Thursday, we build our first comparison rule: how to compare money paid at different times.

Further Reading

These readings are optional and will not appear on a problem set. Use them if you want more background on the Federal Reserve, monetary policy, or the Treasury.

- *The Structure and Functions of the Federal Reserve System*, Federal Reserve Board: who does what inside the Federal Reserve.
- *Economy at a Glance: Policy Rate*, Federal Reserve Board: how the Fed reports and carries out interest-rate policy.
- *Fedpoints: Federal Reserve System*, Federal Reserve Bank of New York: a short overview of how the system is organized.
- *Role of the Treasury*, U.S. Department of the Treasury: what the Treasury does and how it differs from the Federal Reserve.